

Keywords:
scRNA-Seq

50,000+
studies

PubMed

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Drug treatment?
scRNA-seq?
Data available?

No → Delete

Yes →

E-MATB-10698



Download data from
GEO/EMBL-EBI

Gene expression
matrices?

No → Delete

Yes →

Cells annotated?

No → Cells annotation

Yes →

Test cases annotated with key genes

cell	drug	key genes	dose	loc
PC9	erlotinib	["ATF4", "DDIT3", "PPP1R15A", "SLC7A5"]	1uM	UP Fig. 3e, Fig. 4b

Control and perturbed
expression matrices

Manual Data Cleaning and Validation

✓ Data available	Person icon	✓ Case filtering
✓ Key genes directional consistency	Person icon	✓ Cell filtering
✓ Evidence reliable	Person icon	✓ Gene filtering
		✓ Gene mapping
		✓ Data normalization

93 studies, 181 datasets